



SPARTAN INTERNATIONAL SCHOOL

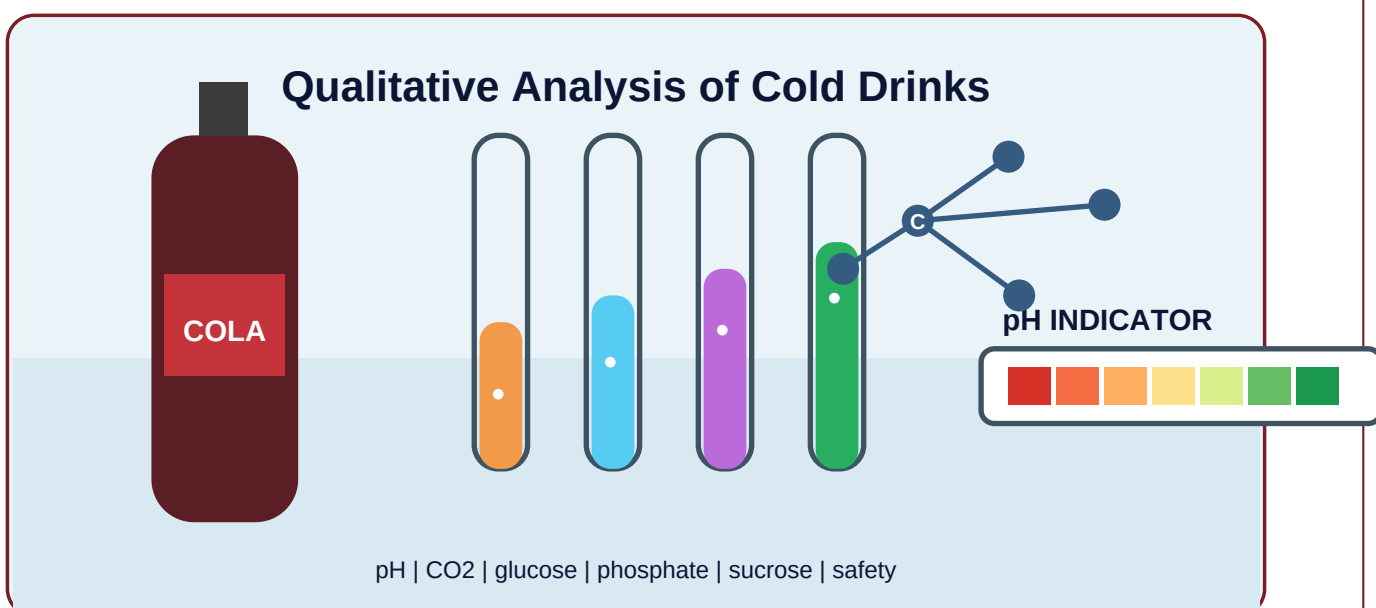
SR. SEC. SCHOOL

INVESTIGATORY PROJECT

ON CHEMISTRY

DETERMINATION OF CONTENTS OF COLD DRINKS

NAME : Joel Jebitto O
CLASS : XII
TOPIC : Cold Drinks Analysis



Acknowledgement

I, **Joel Jebitto O**, a student of Spartan International School, would like to express my sincere gratitude to my Chemistry teacher for guidance and support throughout this investigatory project. The teacher's explanations on qualitative analysis, laboratory precautions and result interpretation helped me complete the work in a systematic manner.

I also thank my school for providing the opportunity to perform and present this project, and my parents for their encouragement. The project helped me understand how common beverages can be examined through simple chemical tests such as pH detection, lime-water test, Benedict's test, ammonium molybdate test, iodoform-type test and heating for sugar residue.

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Project Details

Field	Details
Subject	Chemistry
Class	XII
Topic	Determination of Contents of Cold Drinks
Samples studied	Coca Cola, Sprite, Limca and Fanta

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1. Introduction

Cold drinks are popular carbonated beverages consumed by students, families and the general public. They generally contain water, sweeteners, permitted food acids, flavouring agents, colouring substances and dissolved carbon dioxide gas. The gas produces effervescence and gives a sharp taste when the bottle is opened.

This project investigates selected constituents of commonly available cold drinks through school-level qualitative analysis. The samples selected for comparison are Coca Cola, Sprite, Limca and Fanta. The investigation focuses on acidity, carbon dioxide, reducing sugar, phosphate ions, iodoform-type response and sucrose-related residue formation.

Investigative Focus	Meaning in this project
Acidity	Estimated using pH paper and standard colour comparison.
Carbon dioxide	Detected by passing gas through lime water and observing milkiness.
Glucose / reducing sugar	Detected using Benedict's and Fehling's reagents.
Phosphate	Detected using ammonium molybdate and concentrated nitric acid.
Sucrose	Inferred by heating sample and observing black carbonaceous residue.

2. Purpose and Problem Statement

The purpose of this project is to compare different cold drink samples using qualitative chemical tests. The work does not attempt industrial-grade quantitative analysis. Instead, it demonstrates how basic chemistry can identify the presence or indication of selected constituents.

Cold drinks are often discussed in relation to sugar content, acidity and health impact. A school-level investigation allows the student to connect laboratory techniques with everyday consumer products. It also develops skills in observation, controlled comparison, tabulation and inference.

The problem statement is: **Can commonly available cold drinks be compared through simple chemical tests for acidity, carbon dioxide, glucose, phosphate, alcohol-indicating reaction and sucrose residue?**

Constraint	Treatment in project
Qualitative scope	Results are recorded as present / absent or relative observation.
Brand variation	Four commercial samples are compared under the same test method.
Safety	Strong acids, alkalis and heating are handled only under teacher supervision.
Reliability	Observations are interpreted as indicative, not as precise concentration values.

3. Theory

Cold drinks are acidic mainly because they contain food acids such as citric acid and, in some cola drinks, phosphoric acid. Acidity can be estimated using pH paper. A pH below 7 indicates acidic nature, and a lower pH indicates stronger acidity.

Carbon dioxide dissolves in water and forms carbonic acid. When released and passed through lime water, carbon dioxide reacts with calcium hydroxide to form calcium carbonate, which makes lime water appear milky.

Reaction: $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$

Carbohydrates are important energy-providing compounds. Glucose is a reducing sugar and gives positive reactions with Benedict's reagent and Fehling's solutions after heating. Sucrose is a disaccharide and can leave a black residue on strong heating due to charring of sugar.

Component	Chemical idea	School-level test
Acidic substances	Release H^+ ions in solution	pH paper colour change
Carbon dioxide	Forms carbonate precipitate with lime water	Lime water turns milky
Glucose	Reducing sugar	Brick-red / reddish brown precipitate
Phosphate ions	Formation of ammonium phosphomolybdate	Canary-yellow precipitate
Sucrose / sugar residue	Charring on strong heating	Black residue

4. Chemistry of Cold Drink Components

The project studies reactions or visible changes that are easy to observe in a school laboratory. The following table summarizes the chemical basis of each test and the expected visible change.

Test	Reagent / condition	Expected visible observation	Interpretation
pH detection	pH paper	Pink / orange shade depending on sample	Acidic nature
CO ₂ test	Fresh lime water	Lime water turns milky	Dissolved CO ₂ present
Benedict's test	Benedict's reagent + heat	Reddish precipitate	Reducing sugar indicated
Fehling's test	Fehling A + Fehling B + water bath	Reddish brown precipitate	Reducing sugar indicated
Phosphate test	Ammonium molybdate + conc. HNO ₃ + heat	Canary-yellow precipitate	Phosphate ions indicated
Iodoform-type test	Iodine, KI, NaOH + warm water bath	Yellow precipitate	Positive iodoform-type response
Sucrose residue	Strong heating in china dish	Black residue	Sugar / carbonaceous residue

Note: A school-level qualitative test shows indication of a component or reacting group. For exact composition, instrumental methods such as chromatography, spectroscopy or calibrated titration would be required.

5. Aim, Objectives and Hypothesis

Aim: To perform a comparative qualitative analysis of selected brands of cold drinks available in the market and determine the presence or indication of acidity, carbon dioxide, glucose, phosphate, alcohol-indicating reaction and sucrose residue.

Samples selected: Coca Cola, Sprite, Limca and Fanta.

Objectives:

- To determine the approximate pH of each cold drink sample using pH paper.
- To detect carbon dioxide by passing gas through lime water.
- To test for reducing sugar using Benedict's reagent and Fehling's solutions.
- To identify phosphate ions using ammonium molybdate and concentrated nitric acid.
- To observe an iodoform-type positive response using iodine, potassium iodide and sodium hydroxide.
- To test for sucrose or sugar residue by strong heating in a china dish.

Hypothesis: All selected cold drink samples will show acidic pH and carbon dioxide. Due to added sweeteners and formulation differences, sugar-related observations and relative acidity may vary between brands.

6. Apparatus and Chemicals Required

Apparatus	Use in experiment
Test tubes	Holding small samples and carrying out reactions
Test tube holder and stand	Safe handling and arrangement of heated samples
Beaker	Water bath and solution handling
Burner / spirit lamp	Heating samples under supervision
pH paper	Approximate pH detection
Tripod stand and wire gauze	Support during heating
China dish	Strong heating for sucrose residue test
Stop watch	Recording time for lime-water milkiness

Chemical	Purpose
Lime water	Detection of carbon dioxide
Benedict's reagent	Detection of reducing sugar
Fehling's A and B solutions	Confirmation of reducing sugar
Ammonium molybdate	Detection of phosphate ions
Concentrated nitric acid	Acidic medium for phosphate test
Iodine solution, potassium iodide and sodium hydroxide	Iodoform-type test

7. Detection of pH

Experiment: Small samples of each cold drink were taken in separate clean test tubes. A small strip of pH paper was dipped into each sample. The colour change was compared with the standard pH scale and the approximate pH value was recorded.

Sr. No.	Name of drink	Colour change	Approx. pH value
1	Coca Cola	Pinkish	2 - 3
2	Sprite	Dark orange	3
3	Limca	Light orange	4
4	Fanta	Orange	3 - 4

Inference: The samples are acidic in nature. Coca Cola shows the lowest pH range among the selected samples, while Limca shows a comparatively higher pH value. The variation is due to differences in acid content and formulation.

Investigative note: pH paper gives approximate values only. More accurate values would require a calibrated pH meter.

8. Test for Carbon Dioxide

Experiment: The bottle was opened and the evolved gas from each sample was passed through freshly prepared lime water. The time required for lime water to turn milky was measured using a stop watch.

Principle: Carbon dioxide reacts with calcium hydroxide in lime water to form insoluble calcium carbonate. This precipitate causes the milky appearance.

Reaction: $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$

Sr. No.	Name of drink	Time taken (sec)	Conclusion
1	Coca Cola	28	CO ₂ is present
2	Sprite	20	CO ₂ is present
3	Limca	38	CO ₂ is present
4	Fanta	36	CO ₂ is present

Inference: All selected soft drinks contain dissolved carbon dioxide. The shortest time for milkiess was observed for Sprite, indicating faster CO₂ evolution under this school-level test condition.

9. Test for Glucose

A. Benedict's reagent test: A small amount of cold drink sample was taken in a test tube. A few drops of Benedict's reagent were added and the mixture was heated. Formation of reddish precipitate indicated reducing sugar.

B. Fehling's solution test: Equal amounts of Fehling's A and Fehling's B solutions were added to the sample, and the test tube was heated in a water bath. Reddish brown precipitate confirmed reducing sugar.

Drink	Benedict's observation	Fehling's observation	Conclusion
Coca Cola	Reddish precipitate	Reddish brown precipitate	Glucose / reducing sugar indicated
Sprite	Reddish precipitate	Reddish brown precipitate	Glucose / reducing sugar indicated
Limca	Reddish precipitate	Reddish brown precipitate	Glucose / reducing sugar indicated
Fanta	Reddish precipitate	Reddish brown precipitate	Glucose / reducing sugar indicated

Inference: All selected samples gave a positive reducing sugar indication in this qualitative experiment.

10. Test for Phosphate

Experiment: Small samples of each cold drink were taken in separate test tubes. Ammonium molybdate solution was added, followed by concentrated nitric acid. The mixture was heated carefully under supervision.

Observation principle: The appearance of a canary-yellow precipitate indicates phosphate ions by formation of ammonium phosphomolybdate.

Sr. No.	Name of drink	Observation	Conclusion
1	Coca Cola	Canary-yellow precipitate	Phosphate indicated
2	Sprite	Canary-yellow precipitate	Phosphate indicated
3	Limca	Canary-yellow precipitate	Phosphate indicated
4	Fanta	Canary-yellow precipitate	Phosphate indicated

Inference: The qualitative observations indicate phosphate ions in the selected samples. In commercial drinks, phosphate-related acidity is especially associated with cola-type beverages, while formulation details may vary by manufacturer.

11. Test for Alcohol

Experiment: Small samples of each cold drink were taken in separate test tubes. Iodine solution, potassium iodide and sodium hydroxide solution were added. The test tubes were heated in a hot water bath for about 30 minutes.

Observation: A yellow precipitate was observed in the model experiment, which is treated as a positive iodoform-type response in the school project.

Reaction represented in school project: $\text{CH}_3\text{CH}_2\text{OH} + 4\text{I}_2 + 6\text{NaOH} \rightarrow \text{CHI}_3 + \text{HCOONa} + 5\text{NaI} + 5\text{H}_2\text{O}$

Sr. No.	Name of drink	Observation	Conclusion
1	Coca Cola	Yellow precipitate	Positive iodoform-type response
2	Sprite	Yellow precipitate	Positive iodoform-type response
3	Limca	Yellow precipitate	Positive iodoform-type response
4	Fanta	Yellow precipitate	Positive iodoform-type response

Scientific note: This qualitative test is indicative and may respond to compounds other than ethanol. A confirmatory method would be required before making a precise claim about alcohol content.

12. Test for Sucrose

Experiment: About 5 ml of each cold drink sample was placed in separate china dishes. The samples were heated strongly until a visible residue was formed.

Observation principle: Sugars undergo charring on strong heating and leave a black carbonaceous residue. The intensity of residue gives a rough qualitative indication of relative sugar content.

Sr. No.	Name of drink	Observation	Conclusion
1	Coca Cola	Black residue	Sucrose / sugar residue indicated
2	Sprite	Black residue	Sucrose / sugar residue indicated
3	Limca	Black residue	Sucrose / sugar residue indicated
4	Fanta	Black residue	Sucrose / sugar residue indicated

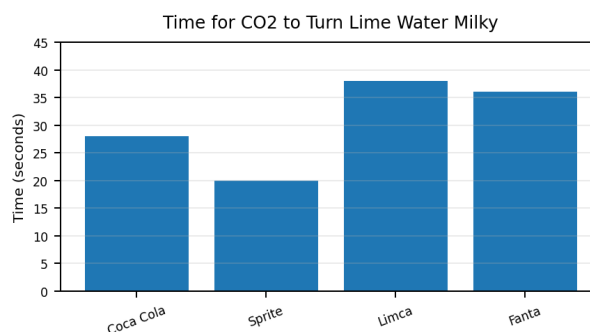
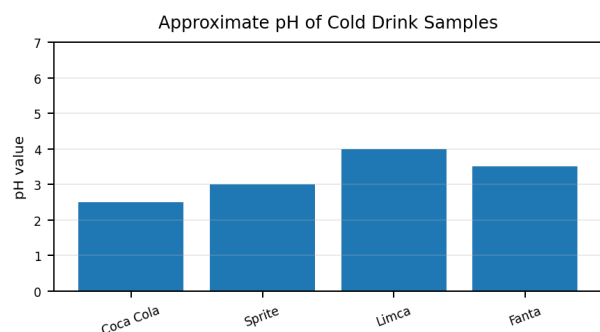
Inference: All brands showed black residue on strong heating. The school-level comparison reports that Fanta showed the strongest sucrose-related residue among the selected samples.

13. Observation Dashboard

The following dashboard summarizes the two semi-quantitative observations recorded during the investigation: approximate pH and time taken for carbon dioxide to turn lime water milky.

Most acidic	Fastest CO ₂ response	Common result
Coca Cola, pH 2 - 3	Sprite, 20 seconds	All samples showed positive qualitative indicators

Drink	Approx. pH	CO ₂ lime-water time	Glucose	Phosphate	Sucrose residue
Coca Cola	2 - 3	28 sec	Positive	Positive	Positive
Sprite	3	20 sec	Positive	Positive	Positive
Limca	4	38 sec	Positive	Positive	Positive
Fanta	3 - 4	36 sec	Positive	Positive	Positive



Interpretation: Lower pH means higher acidity. Shorter CO₂ time indicates faster milkiness formation in lime water, which is used here as a qualitative proxy for CO₂ evolution rate.

14. Result and Comparative Analysis

After conducting the qualitative tests, the selected cold drinks showed the following common pattern: acidic nature, dissolved carbon dioxide, reducing sugar indication, phosphate indication and black residue on strong heating.

Finding	Comparative result
Most acidic sample	Coca Cola, based on the lowest approximate pH range of 2 - 3.
Least acidic sample among selected samples	Limca, based on pH value near 4.
Fastest lime-water milkiness	Sprite, with 20 seconds in the recorded observation.
Slowest lime-water milkiness	Limca, with 38 seconds in the recorded observation.
Reducing sugar indication	All four samples gave positive Benedict's and Fehling's tests.
Sucrose / sugar residue	All four samples produced black residue; Fanta was reported strongest.

Final result: Coca Cola, Sprite, Limca and Fanta all showed positive qualitative indications for the selected constituents tested in this project. The brands differ mainly in relative acidity, CO₂ evolution time and sugar-residue intensity.

15. Precautions and Safety

The following precautions must be followed while performing the investigation:

- Handle concentrated nitric acid and sodium hydroxide only under teacher supervision.
- Wear gloves, lab coat and safety goggles wherever possible.
- Never point a heated test tube toward yourself or another person.
- Use a test tube holder while heating; do not hold hot glassware directly.
- Keep the burner or spirit lamp away from notebooks, paper and loose clothing.
- Do not taste or consume any laboratory sample after chemicals have been added.
- Wash hands thoroughly after completing the experiment.
- If any chemical contacts skin or eyes, wash immediately with plenty of water and inform the teacher.

Risk	Control measure
Acid splash	Use small volumes and keep the test tube away from face.
Heating injury	Use holder, water bath and controlled flame.
Mislabelled samples	Label each test tube before adding reagents.
False inference	Repeat observation and record only what is seen.

16. Health and Consumer Awareness

The investigation shows that cold drinks are chemically active mixtures rather than plain flavoured water. Their acidity, sugar level and dissolved gas content are important from a consumer-awareness perspective.

Aspect	Consumer relevance
Low pH	Frequent intake of acidic beverages can affect dental enamel over time.
Sugar content	High sugar intake may contribute to excess calorie consumption.
Carbonation	Dissolved CO ₂ gives effervescence but may cause discomfort for some people.
Phosphate / acids	Acid additives contribute to taste and preservation but should be consumed moderately.
Labelling	Consumers should read nutrition labels and serving sizes before regular consumption.

Balanced conclusion from consumer awareness: occasional consumption in moderate quantity is different from frequent high intake. Water, fresh fruit juices without excess sugar and balanced diet choices are preferable for daily hydration.

17. Viva Voce Questions

Question	Answer
Why are cold drinks acidic?	They contain food acids such as citric acid and, in some drinks, phosphoric acid.
Which test detects carbon dioxide?	Passing gas through lime water; milkyiness indicates calcium carbonate formation.
What is the reaction in the lime-water test?	$\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$.
Why is Benedict's reagent used?	It detects reducing sugars by forming reddish precipitate on heating.
What indicates phosphate ions?	Canary-yellow precipitate with ammonium molybdate in nitric acid medium.
Why is pH paper not fully precise?	It gives approximate pH by colour comparison, not an exact instrument reading.
What does black residue on heating suggest?	Charring of sugar or organic material.
Why are observations qualitative?	They show presence or indication, not exact concentration.

18. Student Survey Methodology

To connect laboratory observations with daily life, a short student survey can be conducted among 25 to 40 classmates or family members. The survey should not collect private personal data. Responses can be anonymous and used only for this school project.

Proposed questions:

- Which cold drink do you consume most often?
- How many times do you drink cold drinks in a month?
- Do you check the nutrition label before drinking?
- Do you know that cold drinks are acidic?
- Are you aware that high sugar intake can affect health?
- Would you prefer water or fresh juice for regular hydration?

Survey point	Example finding
Most commonly consumed drink	Students may mention cola or lemon-lime drinks.
Label awareness	Many students may not read nutrition labels regularly.
Acidity awareness	Some students may know cold drinks are acidic but not know pH values.
Health awareness	Most students may associate cold drinks with sugar content.

19. Conclusion and Bibliography

Conclusion: The qualitative analysis of Coca Cola, Sprite, Limca and Fanta shows that all selected samples are acidic and contain dissolved carbon dioxide. The samples also gave positive school-level observations for reducing sugar, phosphate indication, iodoform-type response and sucrose-related black residue on heating.

Coca Cola was observed as the most acidic sample among the selected drinks, while Limca showed the highest pH value in the recorded pH-paper comparison. Sprite produced the fastest lime-water milkiness in the CO₂ test. The project demonstrates that chemistry can be applied to everyday consumer products through structured observation and careful interpretation.

Bibliography / References:

No.	Source	Use in project
1	iCBSE Chemistry Project: Determination of Contents of Cold Drinks	Core school-level structure, tests, observations and precautions. URL: icbse.com/projects/chemistry-project-on-determination-of-contents-of-cold-drinks-xnq
2	Comprehensive Chemistry Lab Manual - Class XII	Reference for qualitative analysis and safety.
3	NCERT Chemistry practical concepts	Background theory for pH, carbohydrates and qualitative tests.
4	Teacher guidance and laboratory observations	Final experiment planning, safety and interpretation.

Final statement: This investigatory project connects classroom qualitative analysis with real consumer products and demonstrates the importance of careful observation, safety and responsible interpretation.

End of Project